thm:midsequent in proof-theory/cut-elimination

The Open Logic Project contributors

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Editorial note. All mathematical statement and proof text is reproduced verbatim from the Open Logic Project, unit proof-theory, file midsequent.tex, fixed at commit 1e960beff9ed7835bf3e3f1335e21af3439cd107. Only the packaging (the adapter preamble and the standalone wrapper) is this module's.

Editorial note. Numbering inside this standalone excerpt is editorial, not the Open Logic Project print numbering.

Editorial note. The adapter preamble loads the project's OWN style layer verbatim (sty/open-logic-formulas.sty, sty/open-logic-tokenize.sty, sty/open-logic-selective.sty and open-logic-config.sty, including its tag truth assignment), so the author's text-token shorthand, !A math metavariables, rule names (Intro, Elim, FalseInt) and the tag conditionals resolve exactly as in the book. Three substitutions are forced by the pinned toolchain and are recorded: isomath.sty is absent so \mathsfit is \mathsf; xpunctuate.sty is absent and unused so its RequirePackage is dropped; thmtools/ntheorem are absent so the environments are amsthm with the project's family numbering. The proof trees are the project's own bussproofs-extra.

Editorial note. The H2 judgement and its reason were made in this module by reading the author's own statement and complete proof at the pinned line span. Original candidate key: proof-theory/cut-elimination|midsequent.tex|thm:midsequent.

Editorial note. Reference labels in the proof resolve to navigation pointers to the original author source; these pointers do not reproduce proofs of the prerequisites. The author's own ollabel is kept unprefixed (not part:chapter:section) so those pointers resolve; the label text itself is the author's own argument.

Allowed standard prerequisites (editorial).

- Prerequisites named by the author inside the counted span. The counted span is reproduced verbatim, including the author's own olref pointers. Its prerequisites are the earlier results of the same pinned Open Logic Project commit that the author cites there.

Theorem 0.1 (Midsequent Theorem)**.** *If there is a* **G3c***-derivation* π *of* $\Gamma \Rightarrow \Delta$ *where all formulas in* Γ *and* Δ *are prenex, then there is a derivation* π' *containing a sequent* $\Pi \Rightarrow \Lambda$ *such that all inferences below it are quantifier inferences and all inferences above it are propositional inferences.*

Proof. Define the *order* of a quantifier inference in π as the number of propositional inferences below it, and the order $o(\pi)$ of π as the sum of the orders of all its quantifier inferences. If $o(\pi) = 0$, then no quantifier inference has a propositional inference below it, and we are done: Since all quantifier inferences have one premise, there is a single topmost quantifier inference. Its premise is the midsequent.

If $o(\pi) > 0$, pick a bottom-most quantifier inference with order > 0. Since it has order > 0, there must be a propositional inference below it. Since it is a bottom-most one, its conclusion can't be the premise of a quantifier inference: that second, lower quantifier inference would also have a propositional inference below it otherwise. We distinguish (many!) cases according to what quantifier inference we have, and what the propositional inference below it is. Because the end-sequent consists of prenex formulas only, the principal formula of the quantifier inference can't be active in the following propositional inference. Otherwise the principal formula of that inference would have a quantifier in the scope of a propositional connective. By the sub-formula property, this would have to be a sub-formula of a formula in the end-sequent. So, the principal formula of the quantifier inference is an element of the context of the lower propositional inference.

Here are two examples:

First, suppose we have a $\forall$R which ends in the right premise of $\to$L. Then the relevant sub-derivation of π ends in the sub-derivation π_1.

$$\dfrac{\begin{array}{cc}\vdots\,\pi_2 & \vdots\,\pi_3 \\ \Gamma' \Rightarrow \Delta', \forall x\,\varphi(x), \psi & \dfrac{\chi, \Gamma' \Rightarrow \Delta', \varphi(c)}{\chi, \Gamma' \Rightarrow \Delta', \forall x\,\varphi(x)}\,\forall\mathrm{R}\end{array}}{\psi \to \chi, \Gamma' \Rightarrow \Delta', \forall x\,\varphi(x)}\,{\to}\mathrm{L}$$

We can assume that π is regular, and so the eigenvariable c does not occur outside π_3; in particular, it does not occur in $\psi \to \chi$ or in π_2. Now consider the proof π_1':

$$\dfrac{\dfrac{\begin{array}{cc}\vdots\,\pi_2' & \vdots\,\pi_3 \\ \Gamma' \Rightarrow \Delta', \forall x\,\varphi(x), \varphi(c), \psi & \chi, \Gamma' \Rightarrow \Delta', \forall x\,\varphi(x), \varphi(c)\end{array}}{\psi \to \chi, \Gamma' \Rightarrow \Delta', \forall x\,\varphi(x), \varphi(c)}\,{\to}\mathrm{L}}{\psi \to \chi, \Gamma' \Rightarrow \Delta', \forall x\,\varphi(x), \forall x\,\varphi(x)}\,\forall\mathrm{R}$$

where π_2' is obtained from π_2 by weakening with $\varphi(c)$ on the right. (This does not add any inferences, in particular, no quantifier inferences which might affect the order. It also does not violate any eigenvariable conditions in π_2, since no constants in $\varphi(c)$ are used as eigenvariables in π_2.) The eigenvariable condition on $\forall$R is still satisfied, since c does not occur in $\psi \to \chi$.

We invoke the admissibility of contraction to obtain a derivation π_1'' of $\psi \to \chi, \Gamma' \Rightarrow \Delta', \forall x\,\varphi(x)$. The proof of the admissibility of CR for $\forall x\,\varphi(x)$ and the proof of invertibility of $\forall$R used in that proof did not change the order of any inference: it, at most, deleted inferences. Thus, there are no more quantifier inferences in π_1'' than in π_1, and each quantifier inference in π_1'' has the same

number of proposional inferences below it as its corresponding inference in π_1—with the exception of the last $\forall$R: it had a $\to$L below it in π, which we have moved above it in π_1''. Replace π_1 in π by π_1''. The resulting derivation has a lower order, since at least the order of the $\forall$R inference has been reduced by (at least) 1. Now apply the inductive hypothesis.

The cases where the quantifier inference is $\exists$R is even easier, since contraction is not needed and no eigenvariable condition needs to be worried about. E.g., suppose $\exists$R is followed by $\wedge$R (this time as the left premise). The relevant sub-derivation is π_1:

$$\dfrac{\dfrac{\begin{array}{c}\vdots\\ \pi_2\\ \vdots\end{array}}{\dfrac{\Gamma' \Rightarrow \Delta', \exists x\, \varphi(x), \varphi(t), \psi}{\Gamma' \Rightarrow \Delta', \exists x\, \varphi(x), \psi}\,\exists\mathrm{R}} \qquad \begin{array}{c}\vdots\\ \pi_3\\ \vdots\end{array}}{\Gamma' \Rightarrow \Delta', \exists x\, \varphi(x), \psi \wedge \chi}\,\wedge\mathrm{R}$$

Replace π_1 in π by π_1':

$$\dfrac{\begin{array}{c}\vdots\\ \pi_2\\ \vdots\end{array} \qquad \begin{array}{c}\vdots\\ \pi_3'\\ \vdots\end{array}}{\dfrac{\Gamma' \Rightarrow \Delta', \exists x\, \varphi(x), \varphi(t), \psi \wedge \chi}{\Gamma' \Rightarrow \Delta', \exists x\, \varphi(x), \psi \wedge \chi}\,\exists\mathrm{R}}\,\wedge\mathrm{R}$$

where π_3' is obtained from π_3 by weakening with $\varphi(t)$ on the right. This weakening involves merely adding $\varphi(t)$ to the right side of every sequent in π_3, so does not change the order of any quantifier inference. The order of quantifier inferences is the same in π' as in π, except the order of the $\exists$R inference permuted with $\wedge$R has decreased by 1. The inductive hypothesis applies. $\square$